Project Proposal

Second Year Project

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Project Title : MediCare

Student Name	London MetID	College Mail
Kaushal Adhikari (L)	24045899	np05cp4a240392@iic.edu.np
Darpan Ghimire	24045860	np05cp4a240387@iic.edu.np
Siddhant Tamrakar	24046076	np05cp4a20385@iic.edu.np
Mausam Shrestha	24045923	np05cp4a240276@iic.edu.np
Dikshya Rai	24046165	np05cp4a240359@iic.edu.np

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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Introduction to the project

The population has been growing rapidly in the world. Due to the rapid population growth, the no of the resources has been decreasing and the no of people has been increasing. When we go to the places like hospital and clinics, there is a queue of people waiting for their turn for appointment and checkup. Many of us stand in queue and wait for our turn. Some of us turn came after waiting for a certain time and some of us may be told to come tomorrow. In the hospital, first as sick people come for check ups and health care and if they are told to stand in queue then it will be difficult for them as they are sick. Due to this sometimes neither we get our turn for check up and we lost our valuable time also.

As the modern world is going towards digital technology in each and every sectors. The people will get services we could provide them those service digitally. Neither they lost their time nor they have to wait in queue for their turn in checkup. So, we have created our project named MediCare. The main feature of this project is doctor appointment. It helps patients to checks for appointment of the doctor and book him/her and visit in the fixed schedule. It helps to save time of the people and the problem of standing in the hours of queue is also solved digitally. Nowdays, cellphone has become an essential part of our life. Almost everyone has atleast a single mobile phone. The person who can use use mobile phone can use this service MediCare to book for doctor appointment by sitting in their homes. It neither costs any money and charge any fees. It is free of service.

In this project, we can search doctor and book for appointment if s/he is available. The appointment can be reschedule or cancelled according to our need. Doctors can create and share digital prescriptions. Patients can chat for a quick advice form doctors. It also stores patients health records. The MediCare not only helps and provide service to patients it has also helped doctor as fixed no of patients come for checkup for a fixed schedule of time.

2. Problem Statement

As the inflation rate is increasing rapidly in our country. The services have become more expensive then ever. We have to stand in queue from a month to get the service form public hospitals that our government provide. The people are compelled to pay for health service from private sectors. But due to lack of human resource and the growing population, the people can't get service effectively. The have to wait for a long queue to get checkup from a doctor appointment and some of us are compelled to come back to another day due to yesterday que or doctor not being available. The doctors are compelled to work in 4 or 5 hospitals to provide services to the people.

Therefore, these problems creates the need for a service that solves all of these problems digitally which makes our life easy and also saves our times.

3. Objectives

3.1 General Objectives

To develop a user friendly Full stack environment for medical sector that helps patients to book for doctor appointment and improve medical services like searching doctor, chat for quick advice, share prescriptions form doctors and mange health record of patients.

3.2 Specific Objectives

- To build a Full Stack System to helps patients to book for appointment
- To implement a doctor search and filtering system
- To build a chat system for quick advice.
- To create a digital prescription system.
- To manage patients health records.
- To build a user friendly environment for users.

4. Proposed Solution

The solution of the problem that we have stated above is to build a web based system named MediCare that allows patients to create their own profile in the system to maintain privacy and books for doctor appointment that saves the time of people and solves the problem of standing in a queue. S/he can search doctor by specialization, experience and hospital as we can select our required doctor to book for appointment. Patients can also reschedule or cancel appointment according to their need and necessity. Patients can have a chat to the doctor if they need a quick advice for their problem. Doctors can share digital prescription which patients can watch. It also stores the past records of the patients which can be used for any future problems. As this system saves our time and energy. It is the step towards digital world that makes our life easier.

5. Technology stack

Frontend: HTML, CSS and JavsScript

Backend: Django

Database: MySQL

Version Control: Git & GitHub

Project Management Tools: Ms Teams

6. Methodology

We have used Scrum for the development of our project. Scrum is the framework or the structure for management of the project. It helps to develop projects with teamwork and helps to deliver project faster. In this Scrum methodology a big project is broken down into small manageable parts and finish them in short cycles known as sprints. It helps to finish the project faster as it simplifies the project so that the work can be done effectively. It makes easier to manage a big task as the idea and task is simplified.

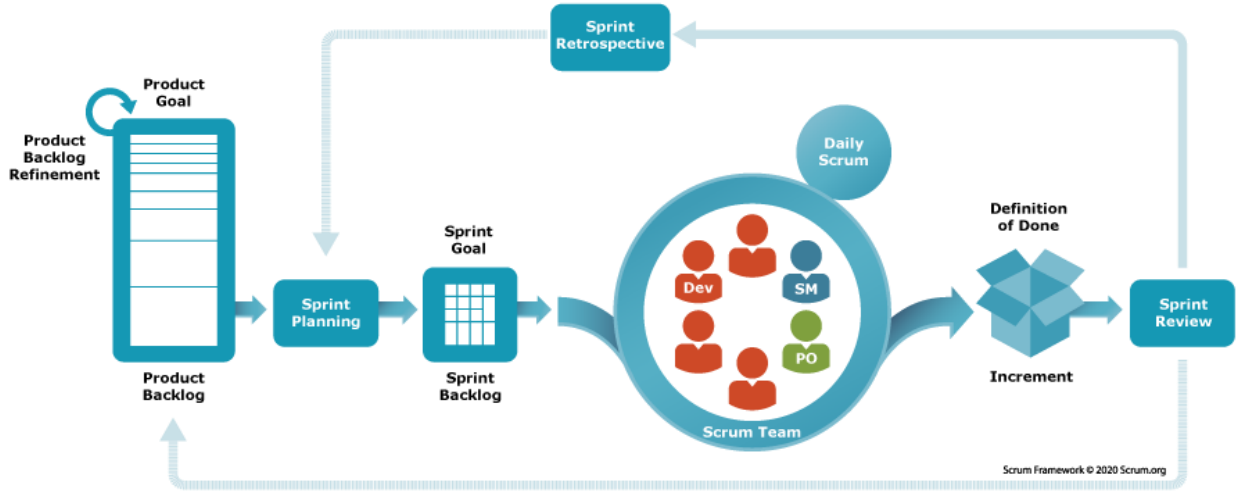

Figure 1: Framework of Scrum

7. Work Breakdown Structure

Work Breakdown Structure also known as WBS is a method in which a big project is broken down into small and manageable tasks that easier to work on big projects. When the big tasks is divided into smaller tasks then big projects can be understand more clearly by the team. So, it helps to improve coordination and communication in the team while doing any project. When we break our tasks, then the tasks can be assigned for team members. It also helps to track progress and reduce confusion and mistakes among group members.

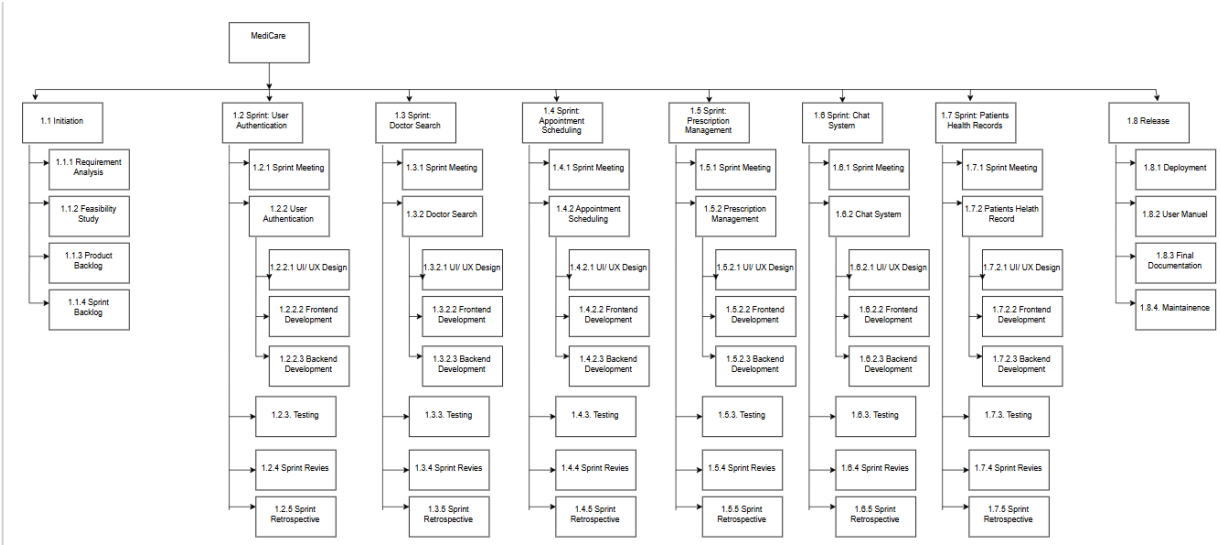

Figure 2:WBS of MediCare

8. ER Diagram

Entity-Relationship Diagram also known as ER diagram is a visual representation of the data and the relationship between them. It is used to plan the data structure.

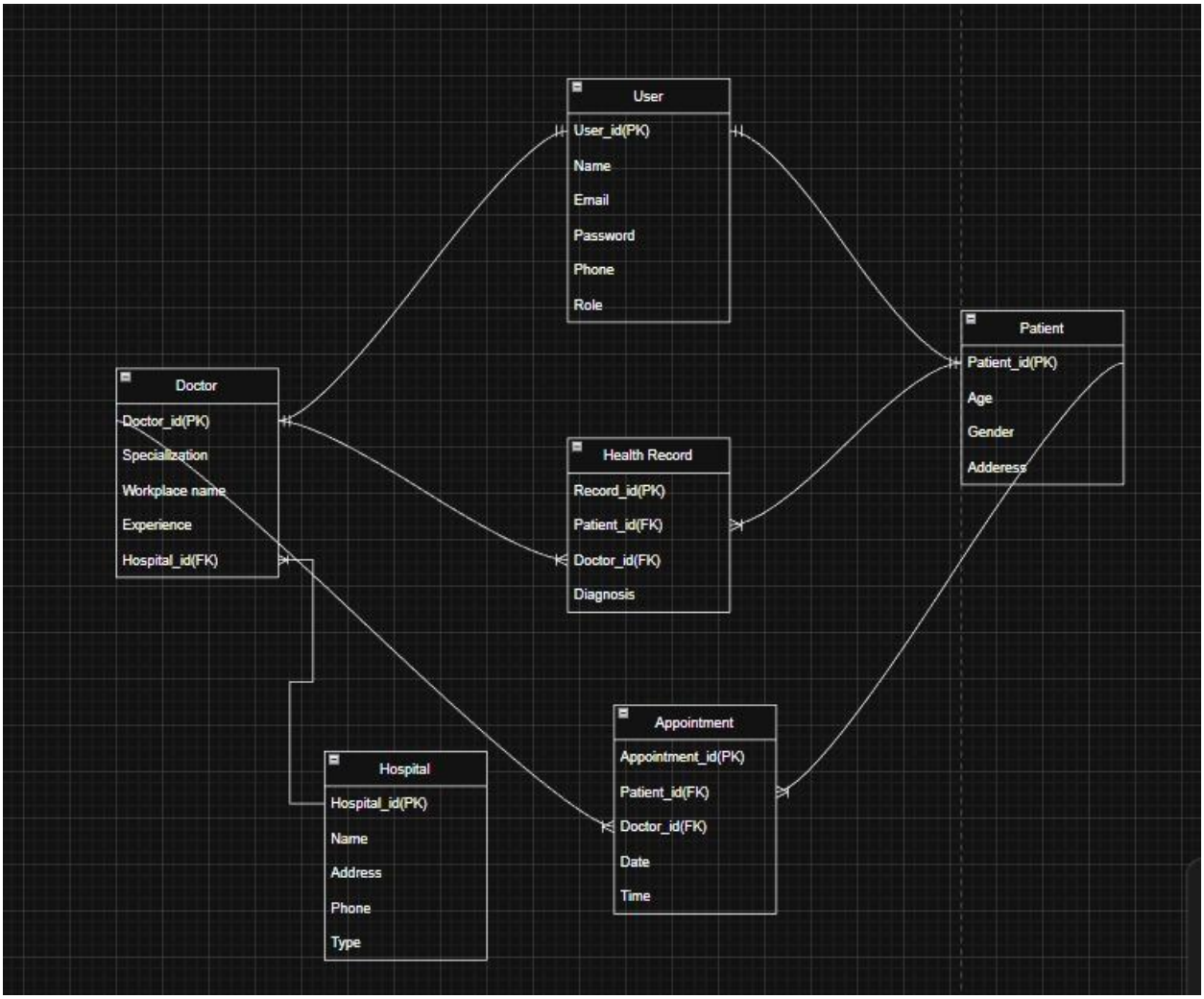

Figure 3: ER Diagram

9. Gantt Chart

Gantt Chart is a project management tool that shows the tasks and the time required for its completion. It helps to simplifies the tasks and time for the tasks. It helps us to know when the task can be completed. It helps us to track the progress of our project.

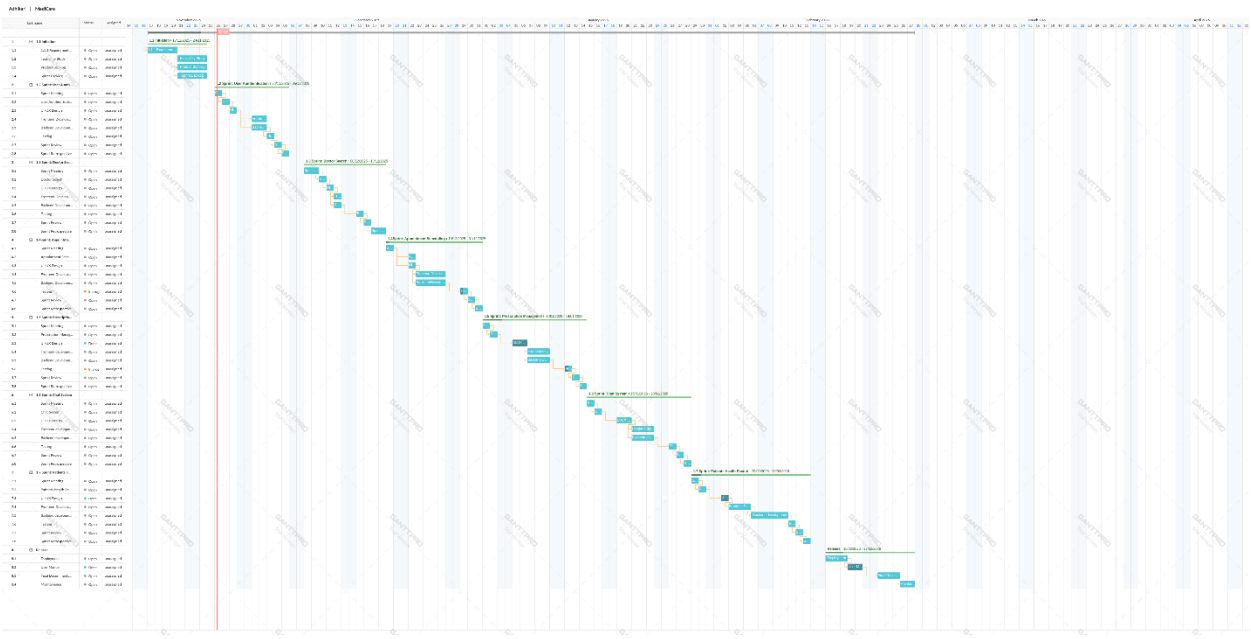

Figure 4: Gantt Chart of MediCare

10. Features and Functionalities

- User login and authentication
- Doctor Search and Filter
- Appointment Scheduling
- Prescription Management
- Chat System
- Patients Health Records

11. Conclusion

The MediCare projects shows us that how digital technology can transform our life. It makes our life easier as it provides services much faster than the traditional healthcare system where we need to stand in queue. It is more reliable and can be access from anywhere with the help of internet. The project provides a solution for both patients and doctors as it provides features such as doctor search, appointment with doctor, prescription management, store patients health record and a chat system.

The system is user friendly as it can be used by anyone who knows how to use a smart phone. The projects helps to reduce in queue in the hospitals as well as save the time of patients. The use of technologies such as HTML, CSS and JavaScript for frontend, Django for backed and My SQL for the database ensures the system is reliable, scalable and secure. We have used the SCRUM methodology for the proper planning and breakdown of the tasks for its proper completion. Overall, our system provides a smooth, accessible and digital healthcare experience in this modern world.